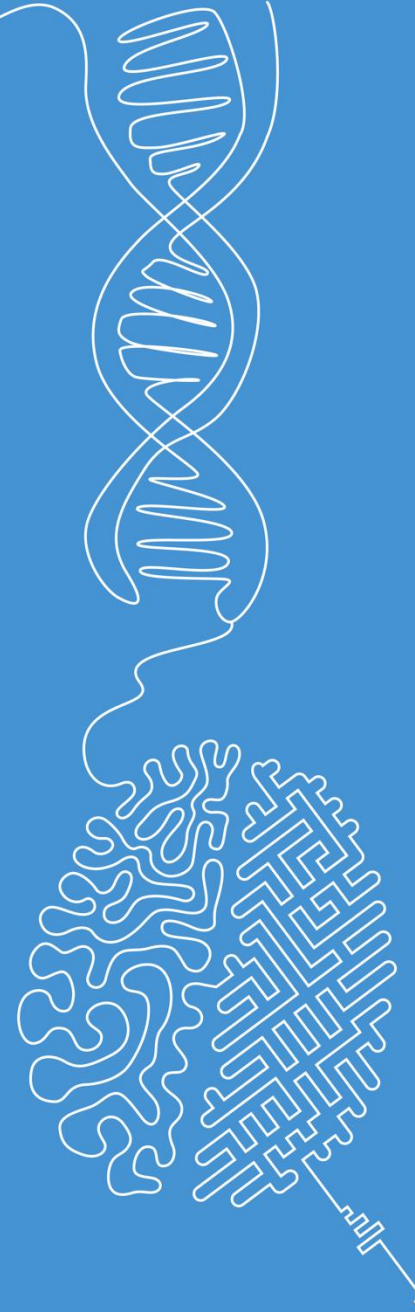


Interpolation

Image and Signal Processing

Norman Juchler

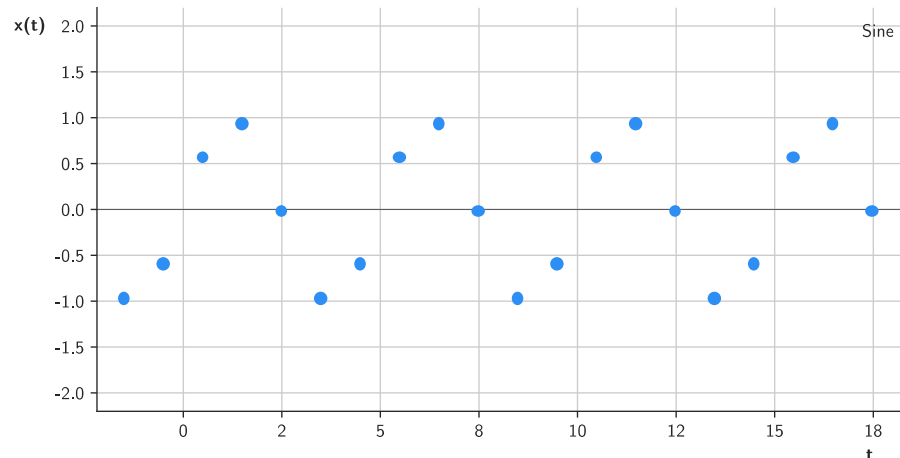


Interpolation

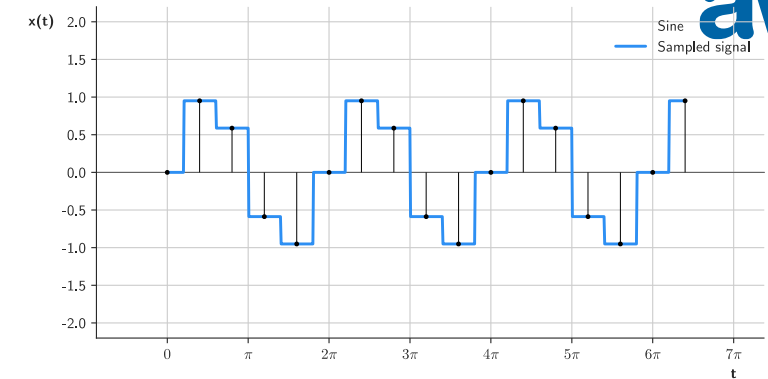
...undo sampling

Interpolation

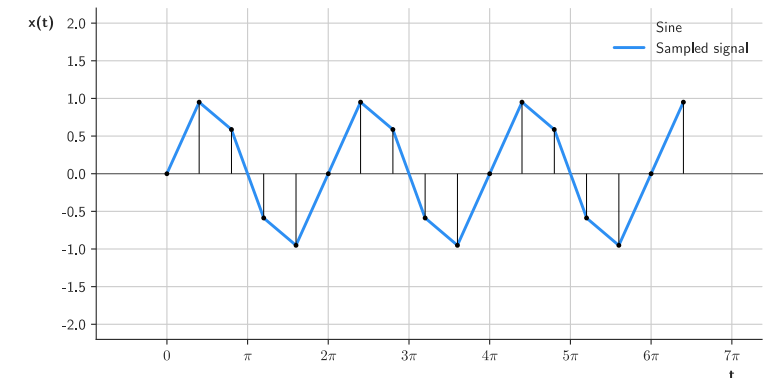
- **Definition:** Interpolation is the process of reconstructing a continuous-time signal $x(t)$ from its discrete samples $x[n] = x(nT_s)$, where T_s represents the sampling interval.
- **Purpose:** Interpolation allows us to estimate the signal values at points between the original samples.
- **Input:**



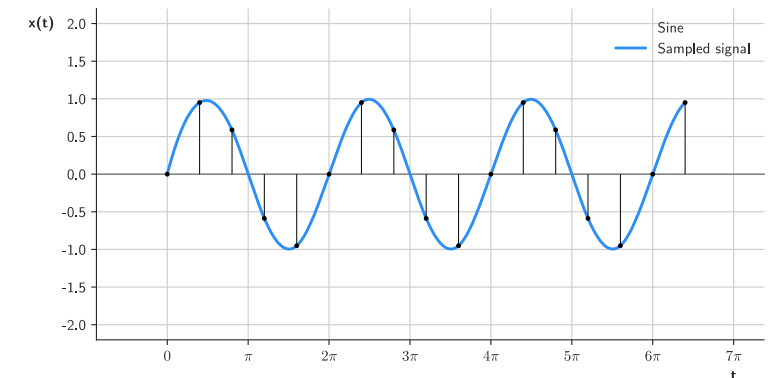
Nearest
neighbor
interpolation



Piecewise
linear
interpolation



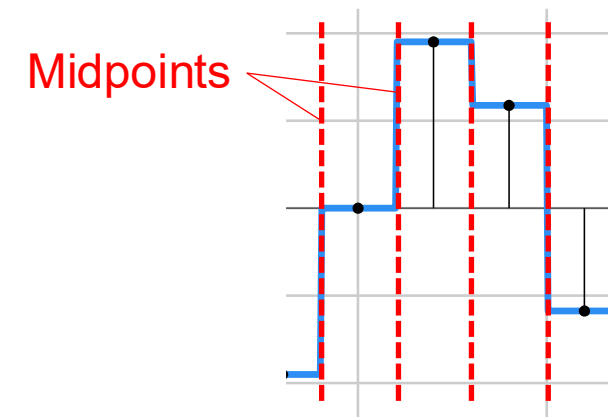
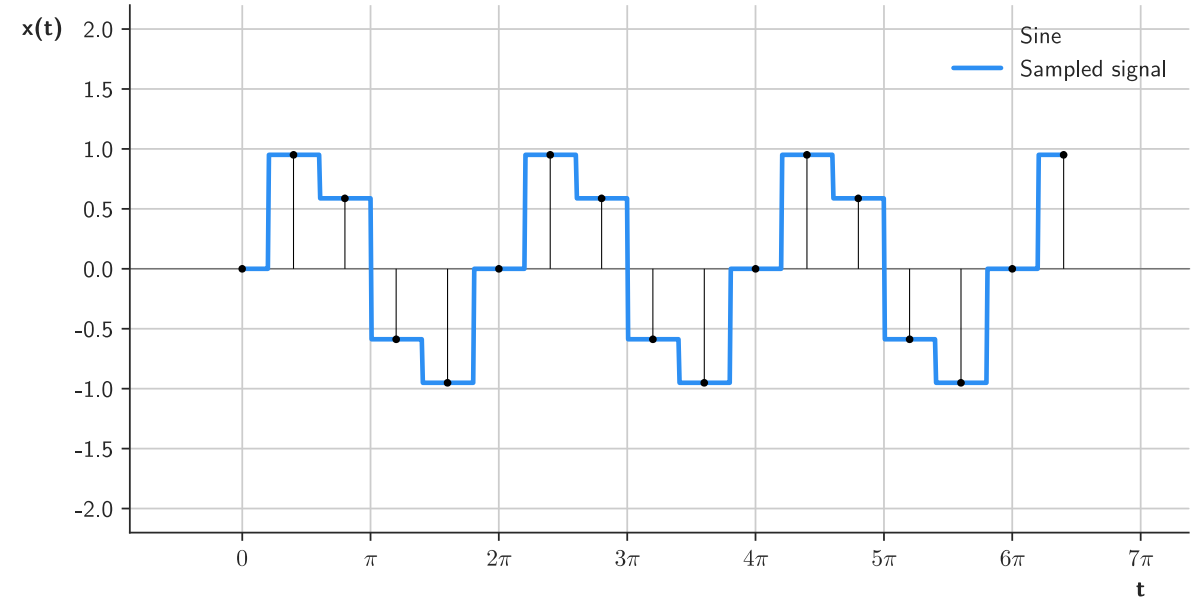
Piecewise
cubic
interpolation



Nearest neighbor interpolation

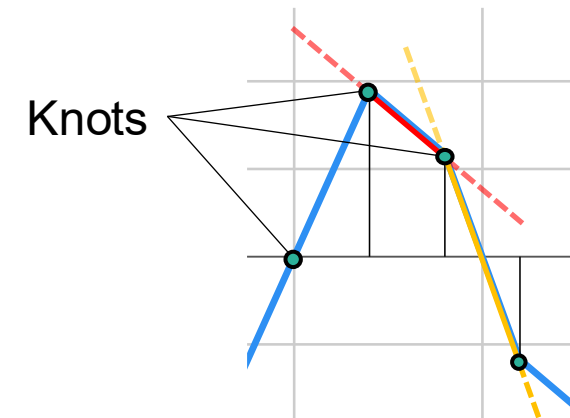
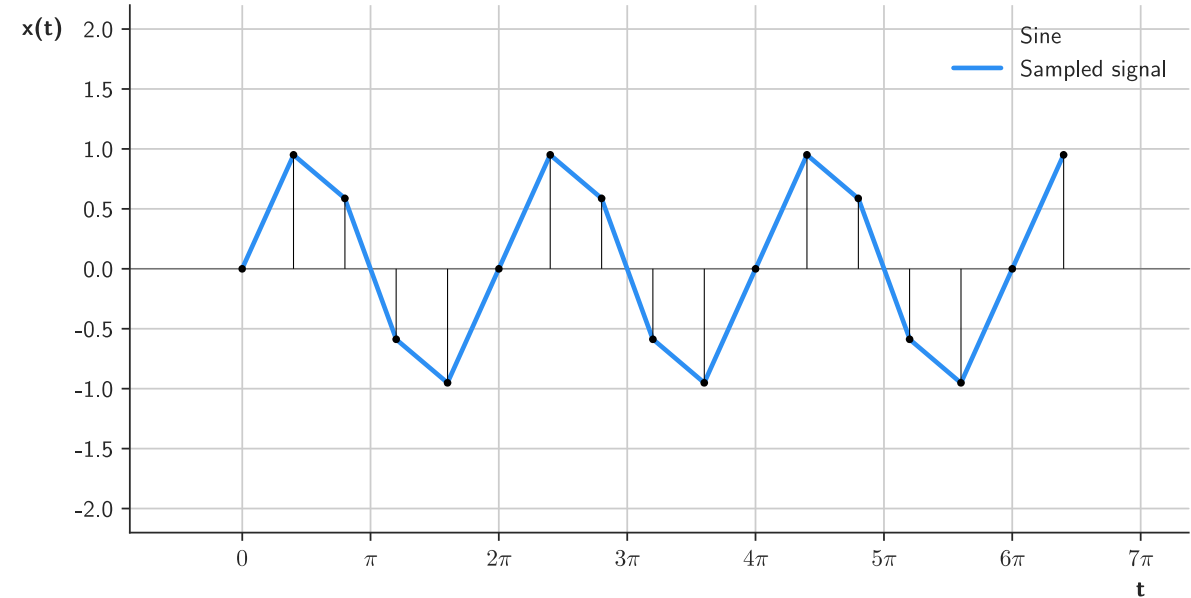
- Given data points $(t[i], x[i])$, the function approximates $x(t)$ by assigning the value of the nearest known data point.
- The interpolated function is piecewise constant between the midpoints:

$$\hat{x}(t) = x[i], \quad \text{where } t \in \left[\frac{t[i-1] + t[i]}{2}, \frac{t[i] + t[i+1]}{2} \right)$$
- Properties:** The function is discontinuous and not smooth at the midpoints



Piecewise linear interpolation

- Given data points $(t[i], x[i])$, the function linearly interpolates between adjacent points.
- Linear segments:
$$\hat{x}(t) = a_1(t - t[i]) + a_0, \quad t[i] \leq t \leq t[i + 1]$$
$$x(t[i]) = x[i]$$
$$x(t[i + 1]) = x[i + 1]$$
- **Properties:** The function is continuous, but not differentiable at the knots $x[i]$.
- **Knots:** Points at which the segments are “glued” together



Piecewise cubic interpolation

- Given data points $(t[i], x[i])$, the function interpolates using a cubic polynomial

- Cubic segments:

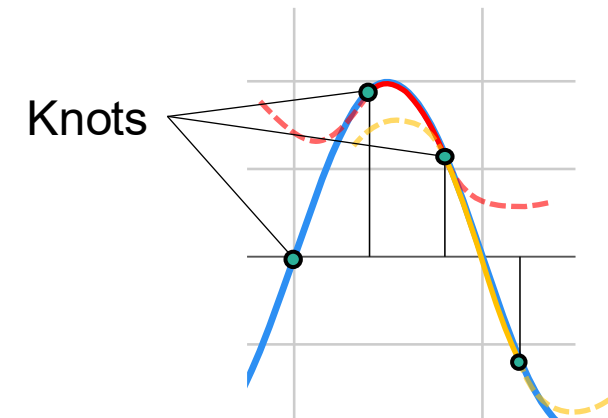
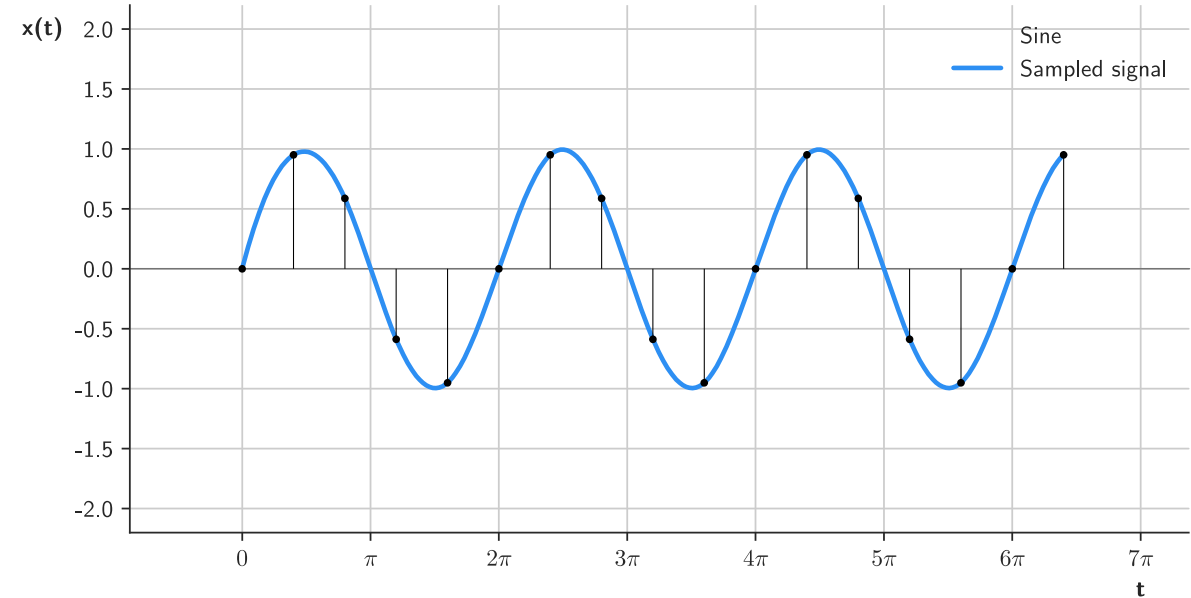
$$\hat{x}(t) = a_3(t - t[i])^3 + a_2(t - t[i])^2 + a_1(t - t[i]) + a_0,$$

$$t[i] \leq t \leq t[i + 1]$$

$$x(t[i]) = x[i], \quad x(t[i + 1]) = x[i + 1],$$

$$x'(t[i]) = \dots, \quad x'(t[i + 1]) = \dots$$

- The coefficients are determined by ensuring continuity and smoothness (e.g., Hermite or spline conditions)



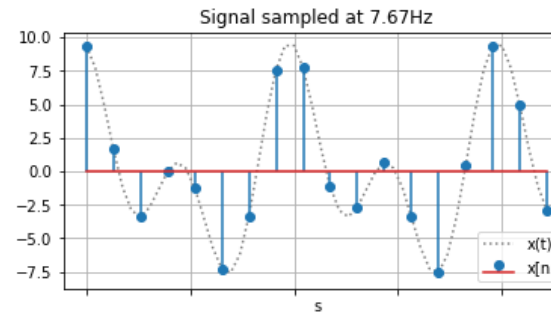
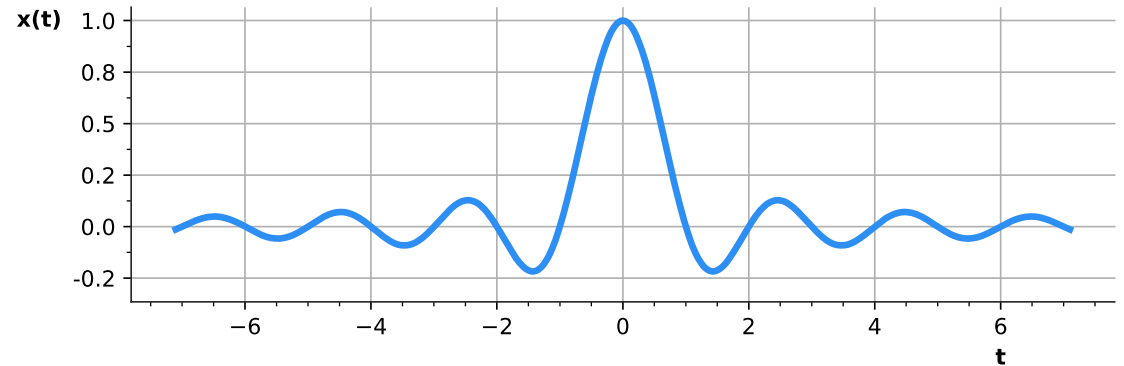
Whittaker–Shannon interpolation (sinc interpolation)

- Given N data points $(t[i], x[i])$ sampled at fixed time intervals T , approximate the function $x(t)$ using

$$\hat{x}(t) = \sum_{i=0}^{N-1} x[i] \operatorname{sinc}\left(\frac{t - t[i]}{T}\right)$$

$$\operatorname{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$$

- The function is differentiable and reconstructs the original signal if sampled correctly (no aliasing).
- Useful for band-limited signals (i.e., if the signal does not contain freq. larger than the Nyquist frequency)



- 1 A continuous-time signal $x(t)$ sampled at a given sampling frequency yields a discrete sequence of samples $x[n]$.

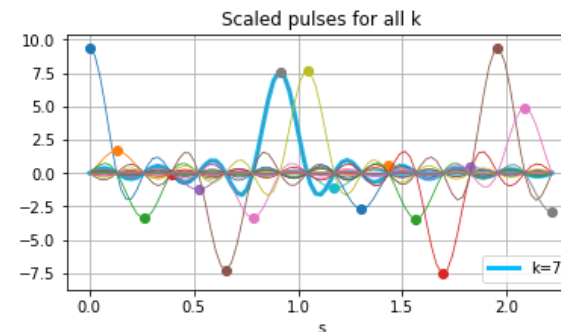
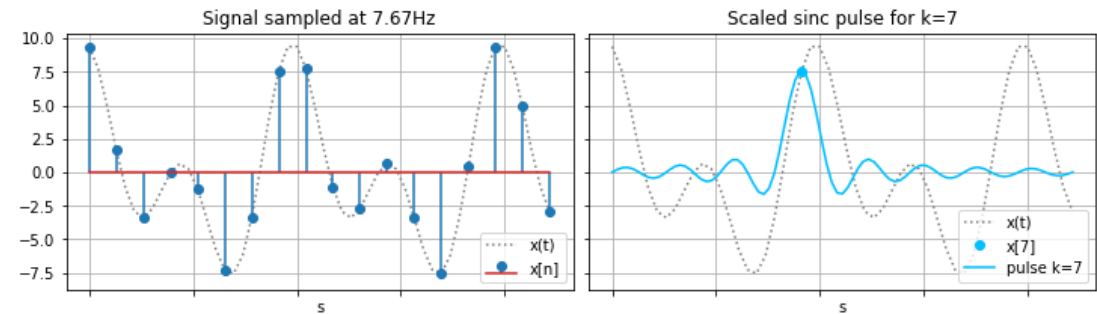
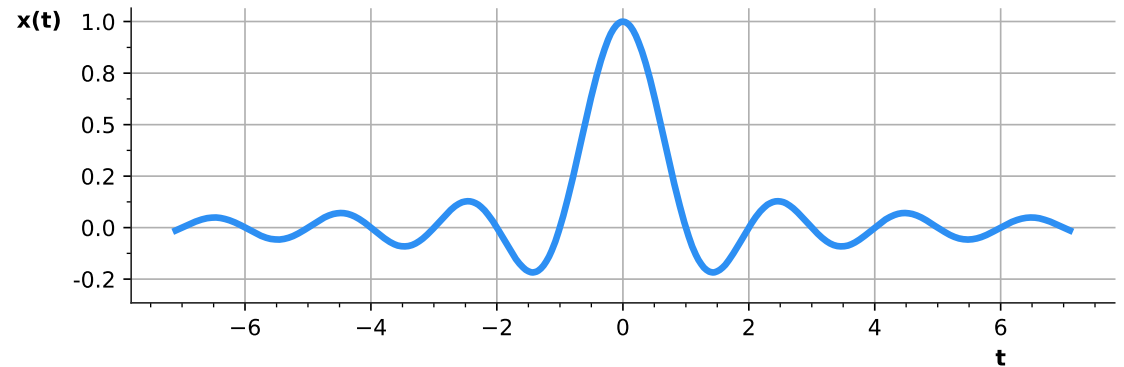
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- For each sample, we construct a shifted and scaled sinc function whose peak aligns with that sample value.

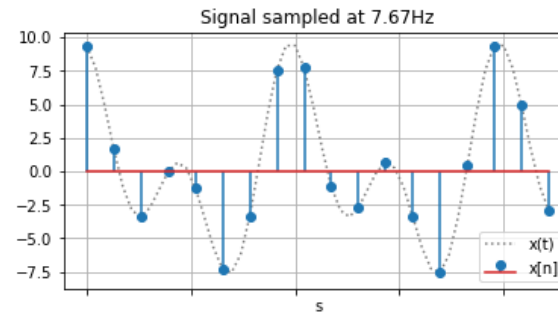
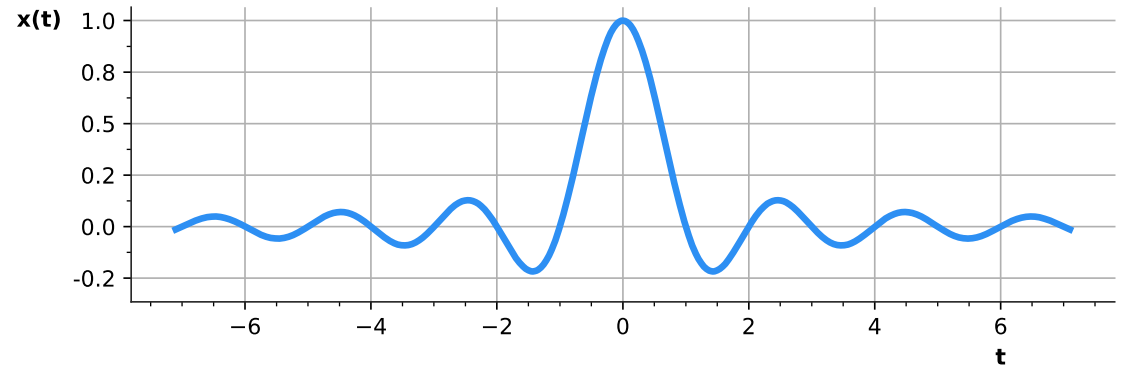
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- ③ Summing all these sinc functions yields a reconstruction of the original signal. This process is known as sinc-interpolation.

